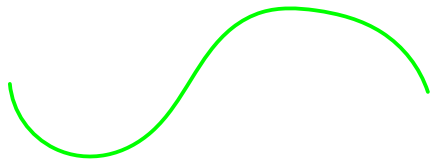
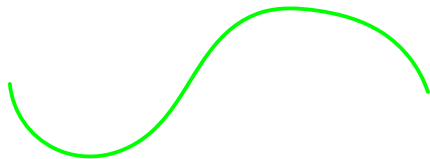


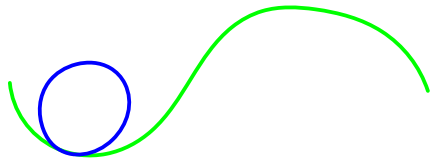
Another view on curvature

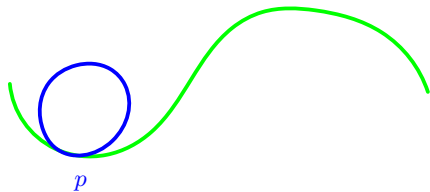


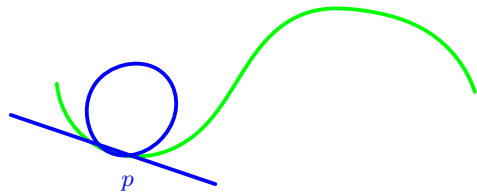
Another view on curvature

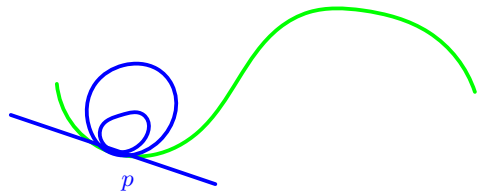


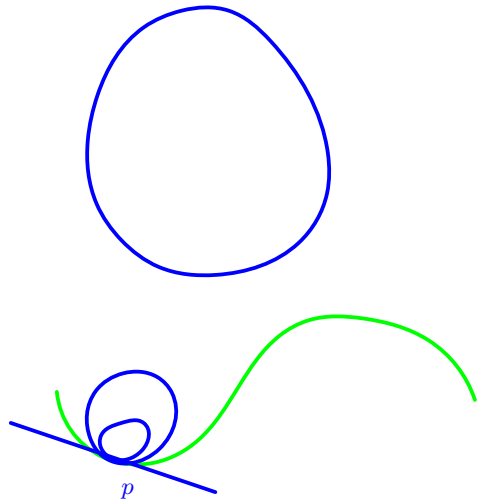
Another view on curvature

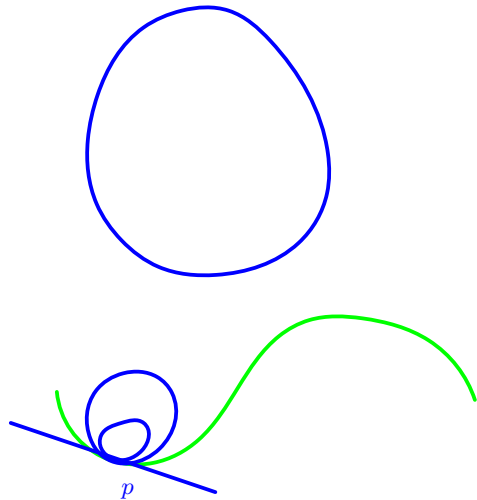


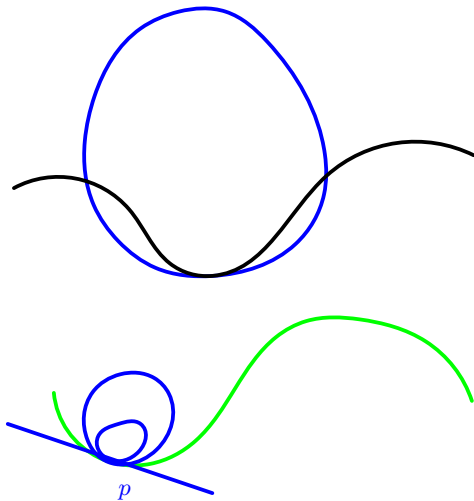


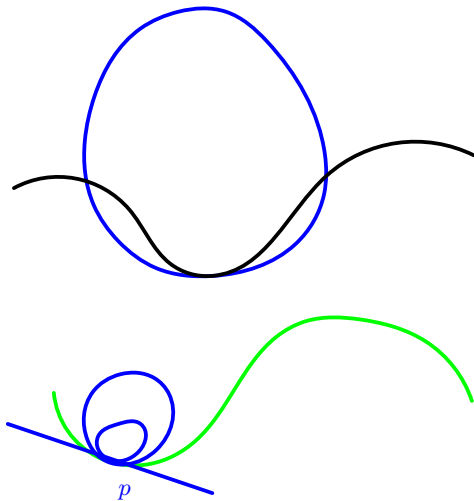


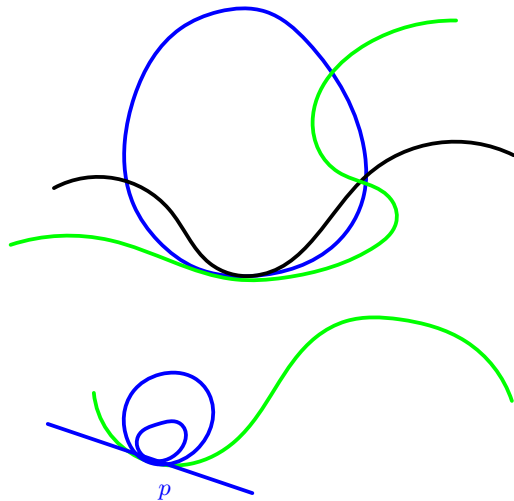


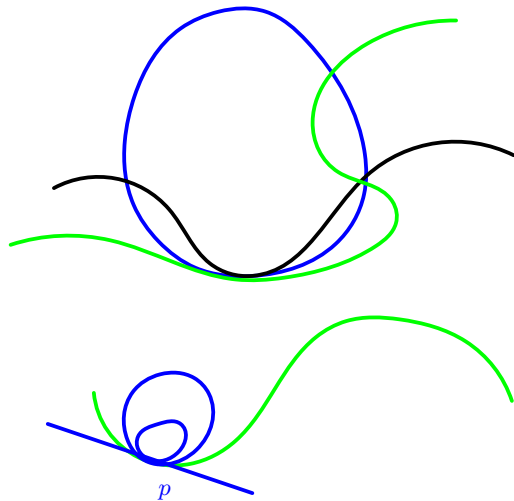


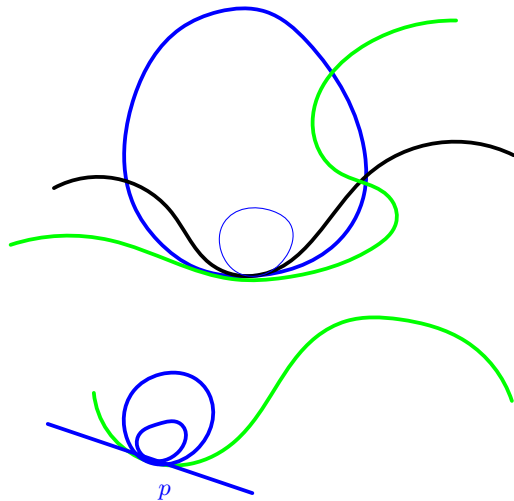


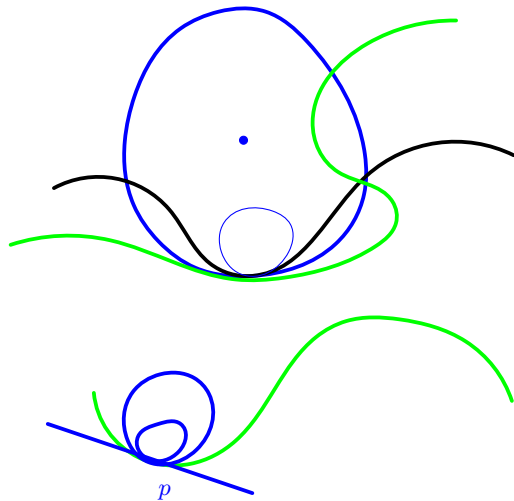


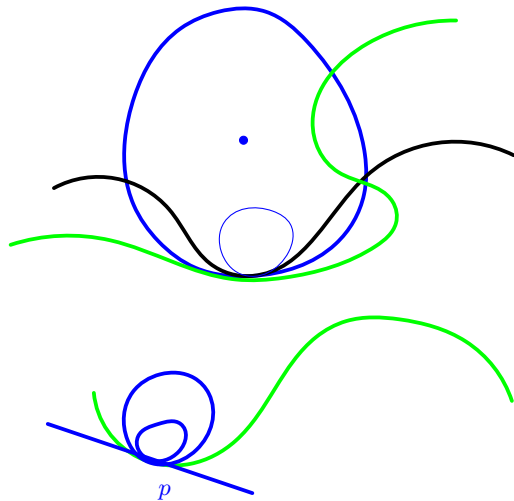


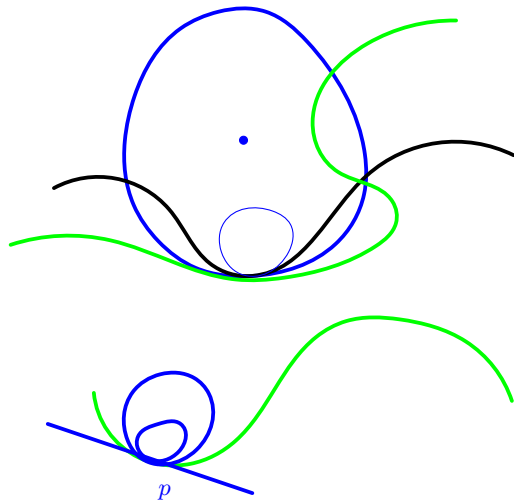


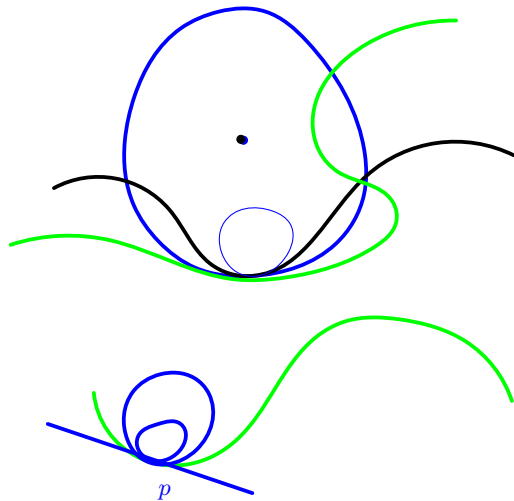


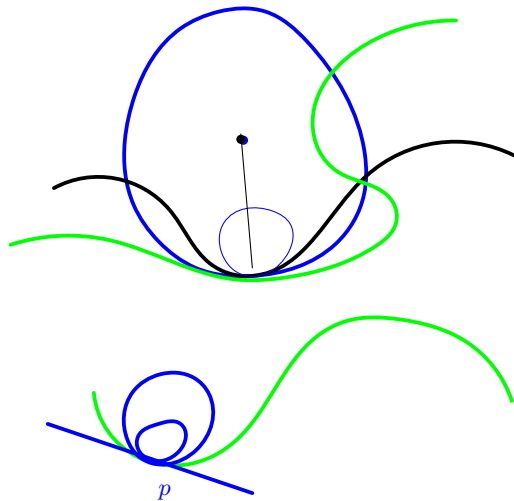


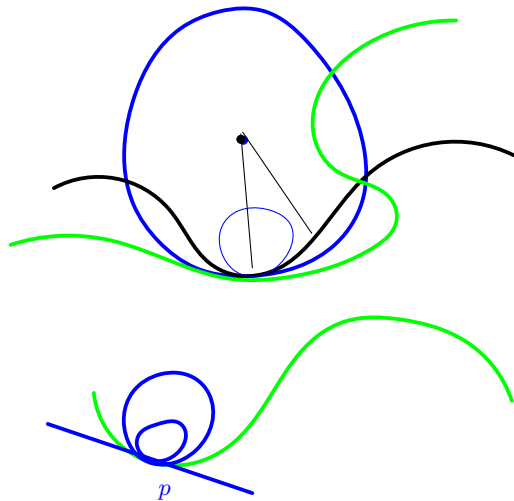


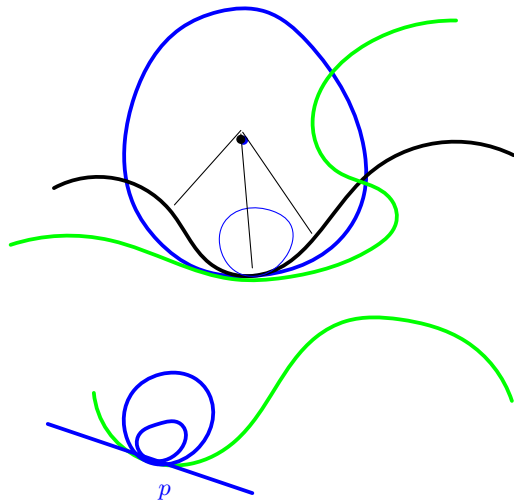


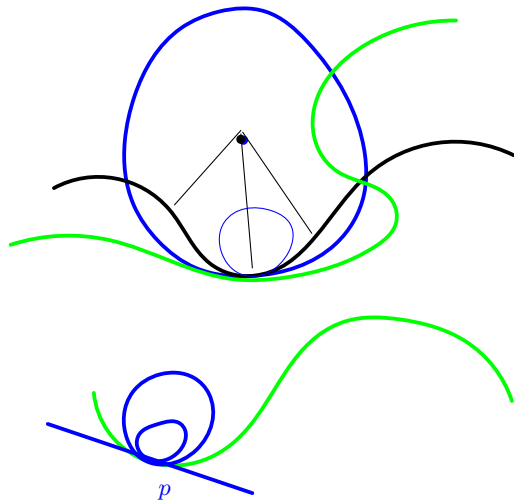


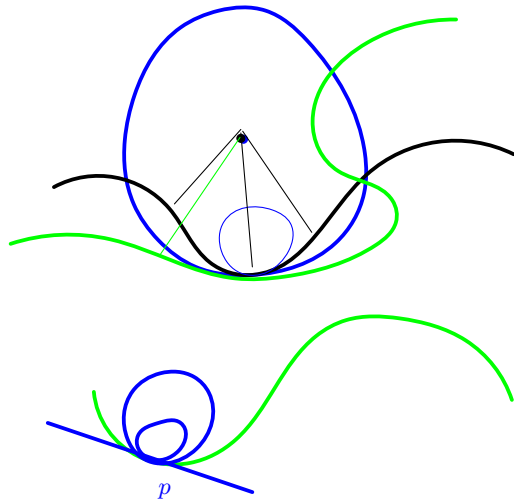


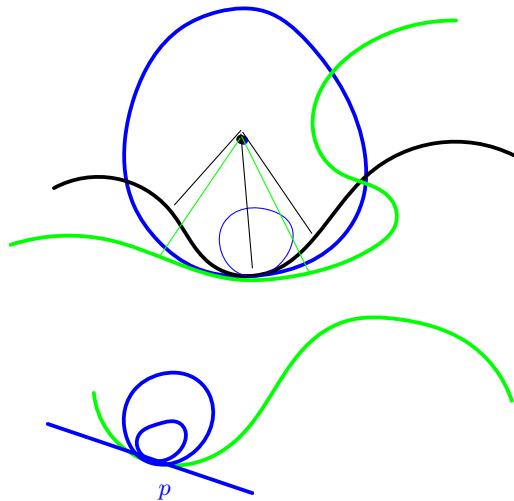


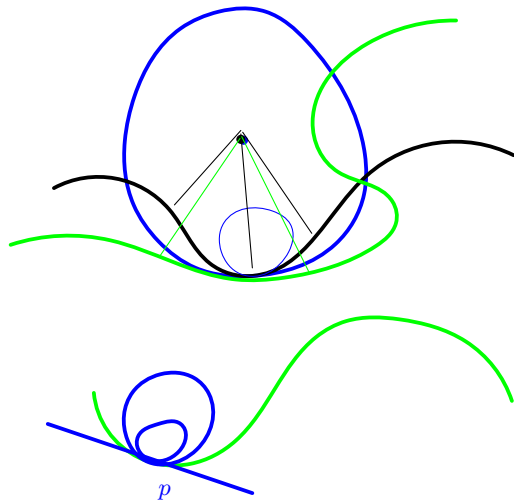






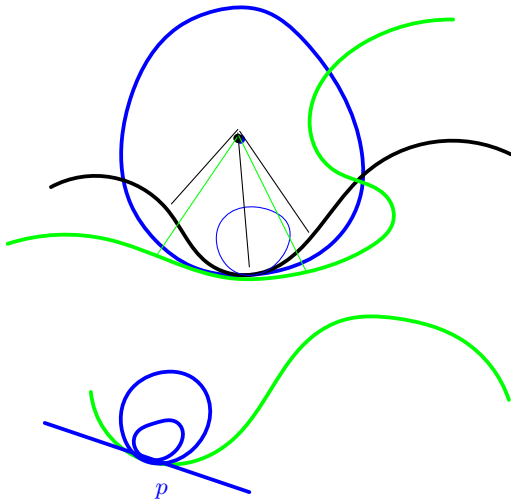






Another view on curvature

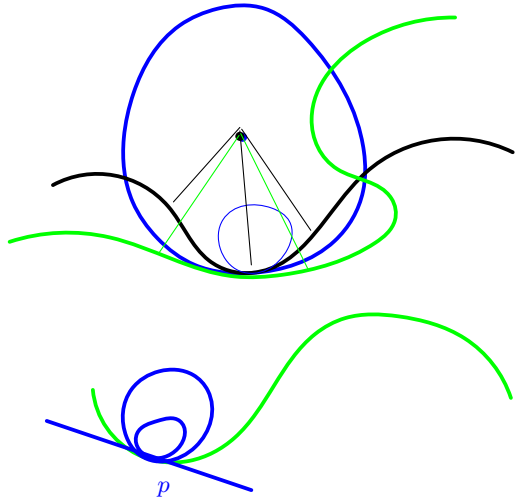
Given $\theta : (a, b) \rightarrow \mathbb{R}^2$



Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

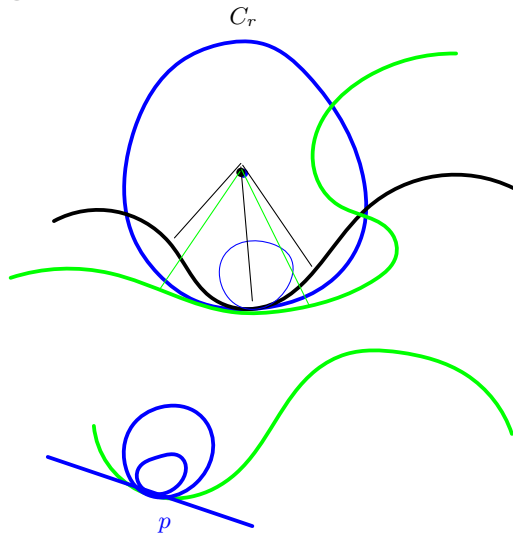


Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

○



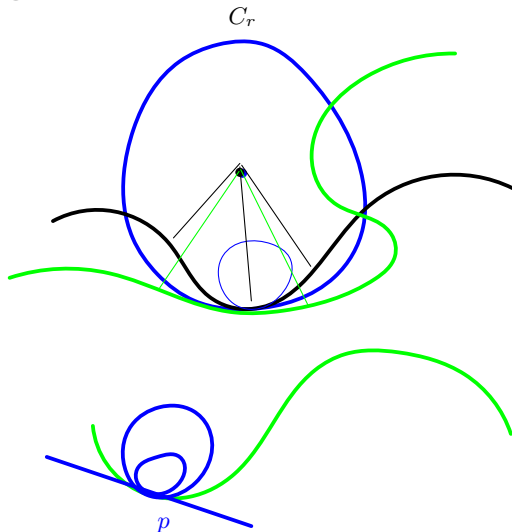
Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

$d(t)$ is the distance of $\theta(t)$ from the center of C_r

○



Another view on curvature

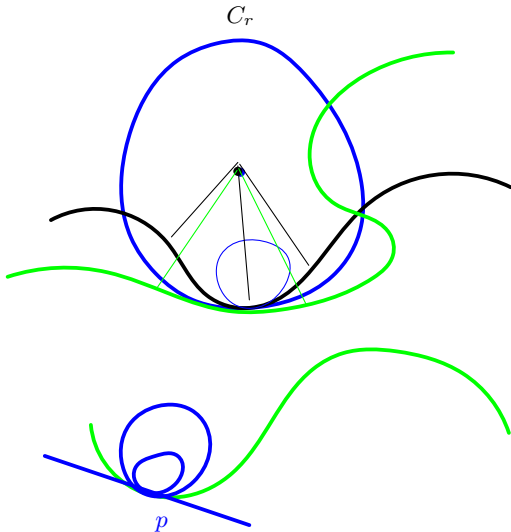
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○



Another view on curvature

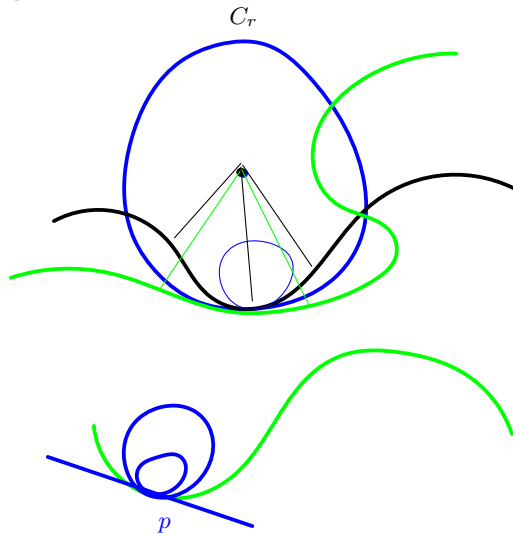
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$d(t)$ is the distance of $\theta(t)$ from the center of C_r

$d(t_0) = r$

○



Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

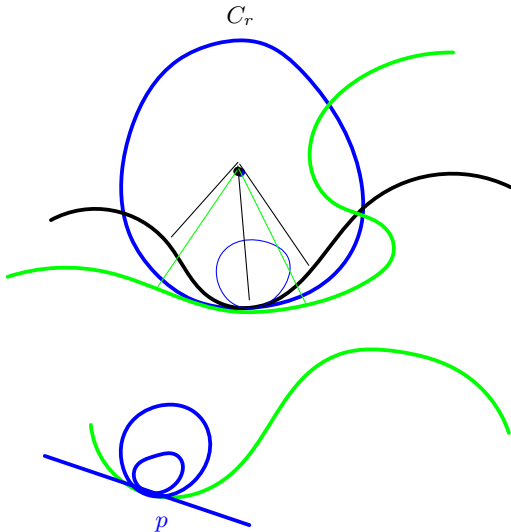
Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

$d(t)$ is the distance of $\theta(t)$ from the center of C_r

$d(t_0) = r$

If the C_r is too big, t_0 is a local maximum for d

○



Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

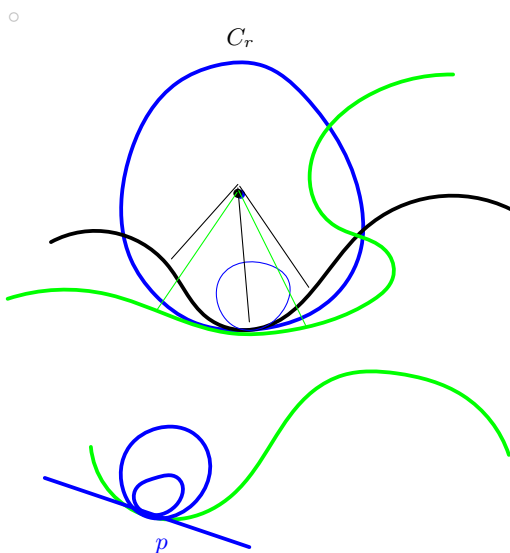
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Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

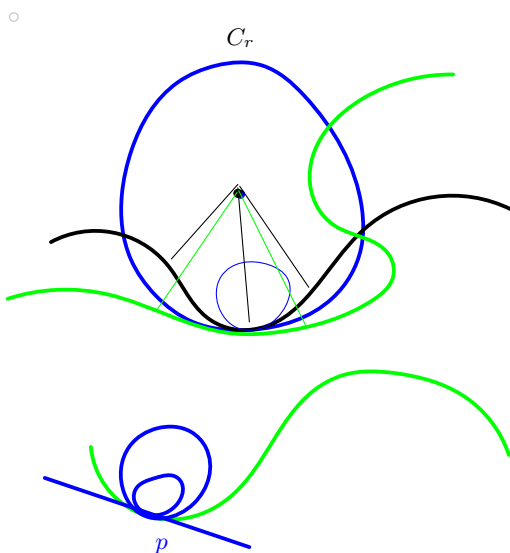
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C_r is too big if $d''(t_0) < 0$



Another view on curvature

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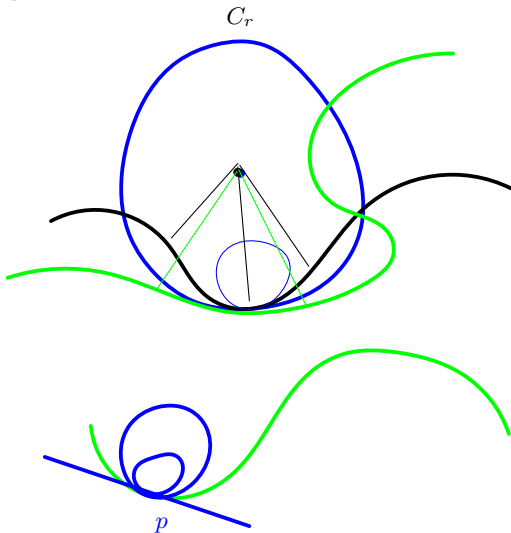
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○



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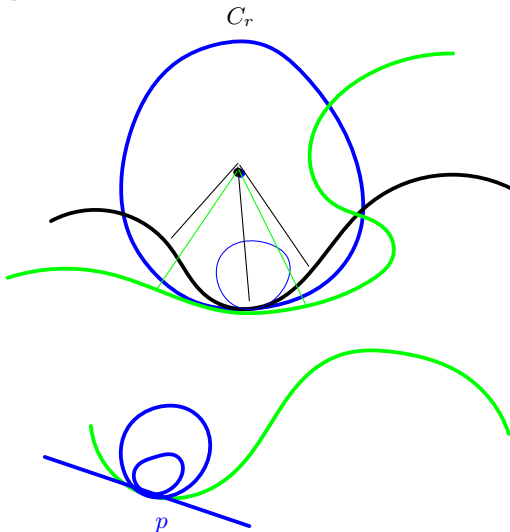
If C_r is too small, t_0 is a local minimum for d

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○



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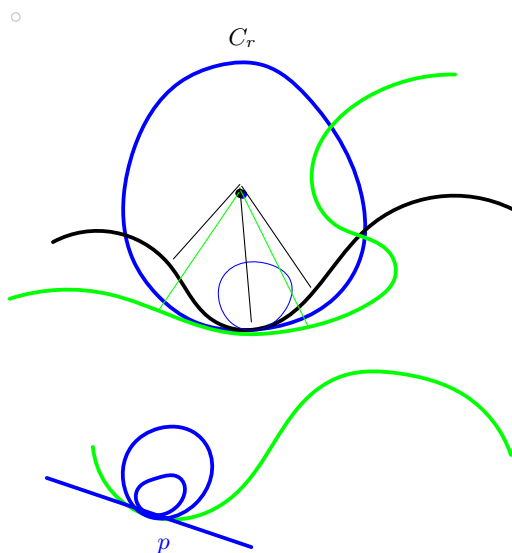
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The curvature will be $1/r$ for that r



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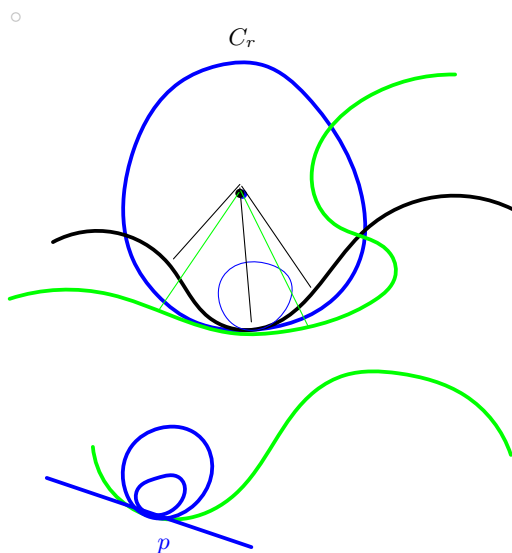
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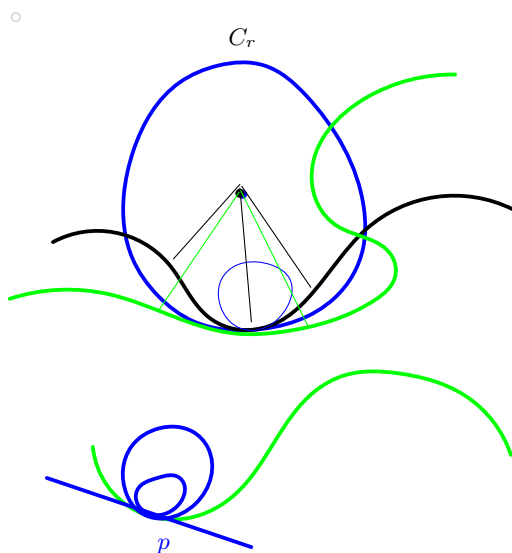
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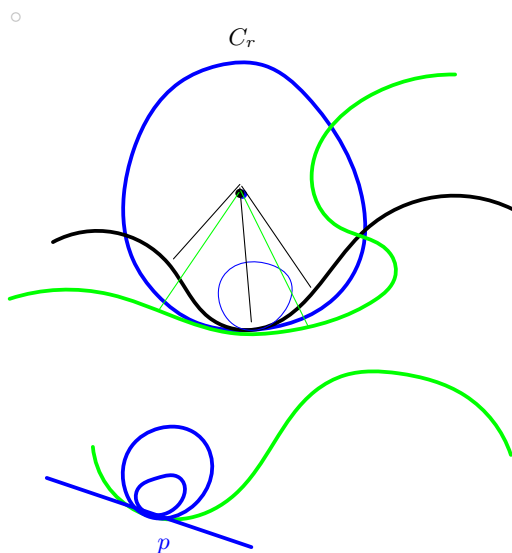
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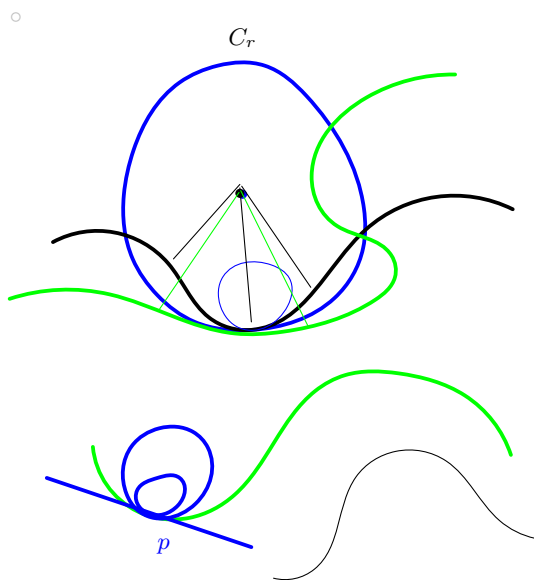
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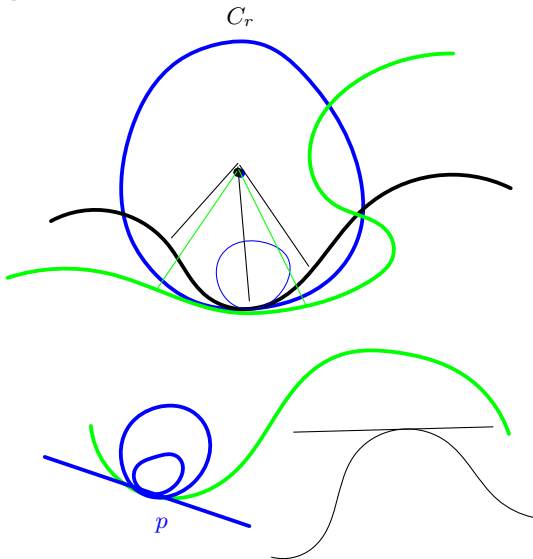
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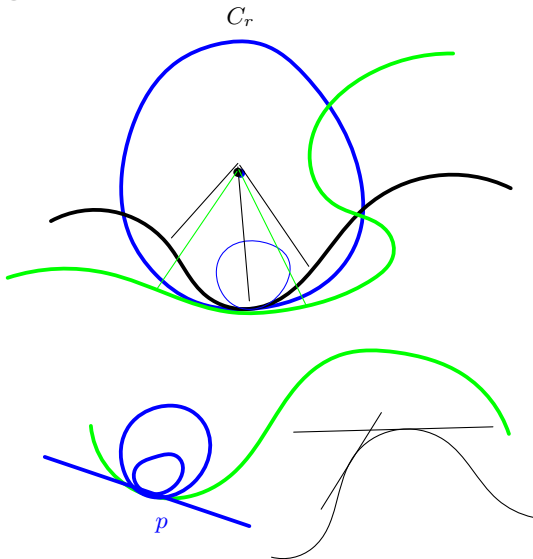
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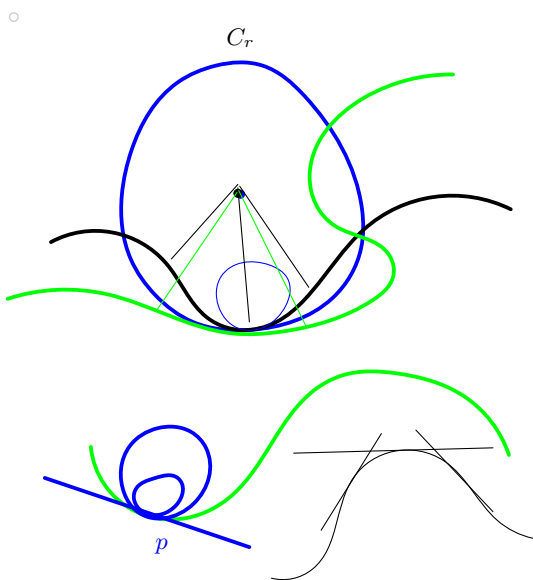
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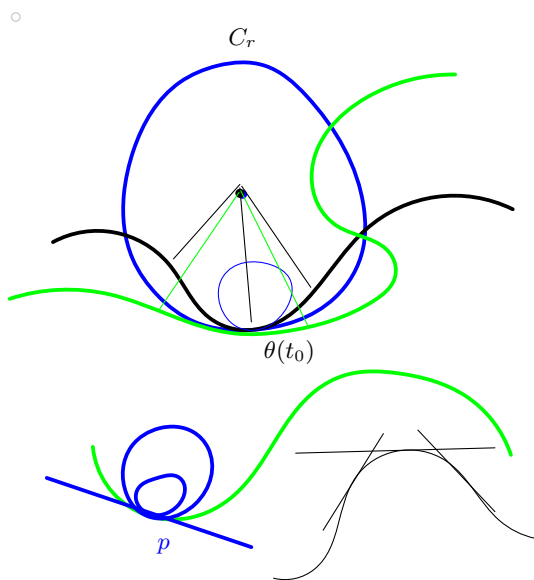
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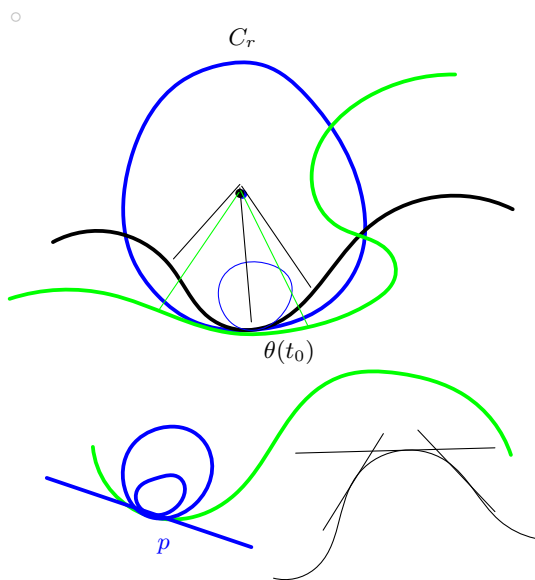
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What is the center of C_r ?



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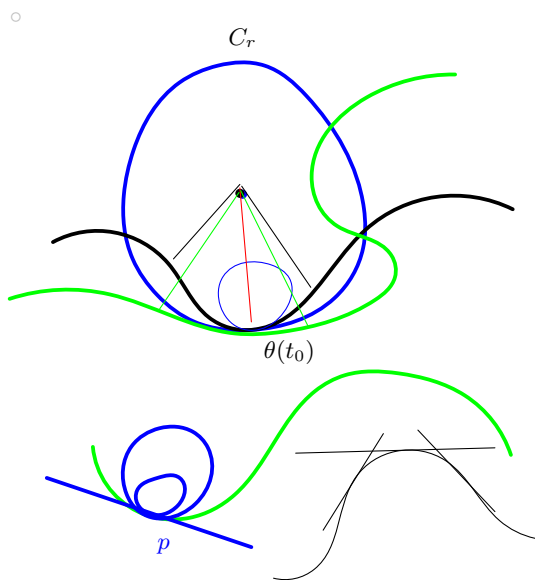
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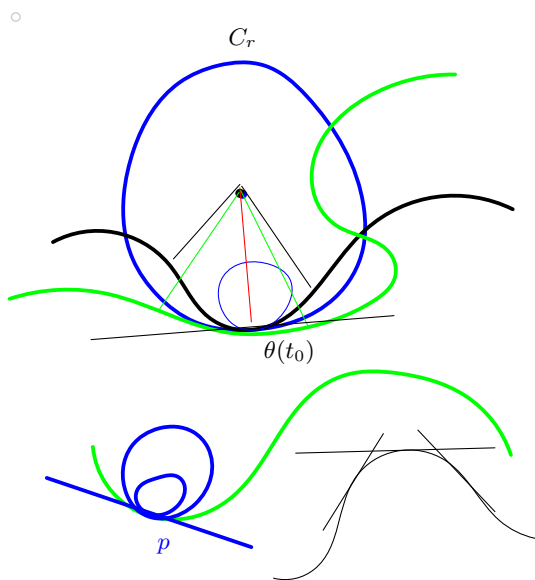
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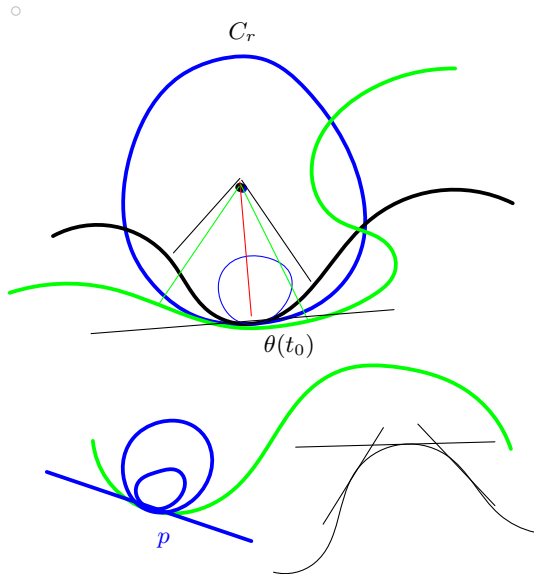
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Center of the circle: $\theta(t_0) + r\mathbf{N}(t)$



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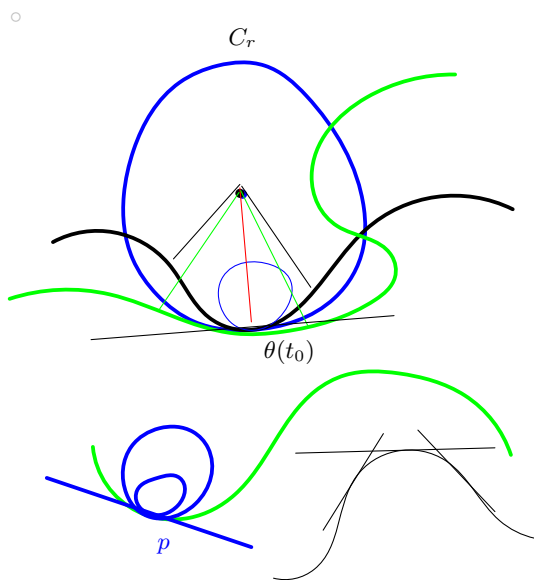
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$d(t) = \|\theta(t) - \theta(t_0) - r\mathbf{N}(t)\|$



Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$

Supposing a circle C_r of radius r is tangent to the curve parametrized by θ at $\theta(t_0)$

$d(t)$ is the distance of $\theta(t)$ from the center of C_r

$d(t_0) = r$

If the C_r is too big, t_0 is a local maximum for d

If C_r is too small, t_0 is a local minimum for d

C_r is too big if $d''(t_0) < 0$

C_r is too small if $d''(t_0) > 0$

C_r is the best fit when $d''(t_0) = 0$

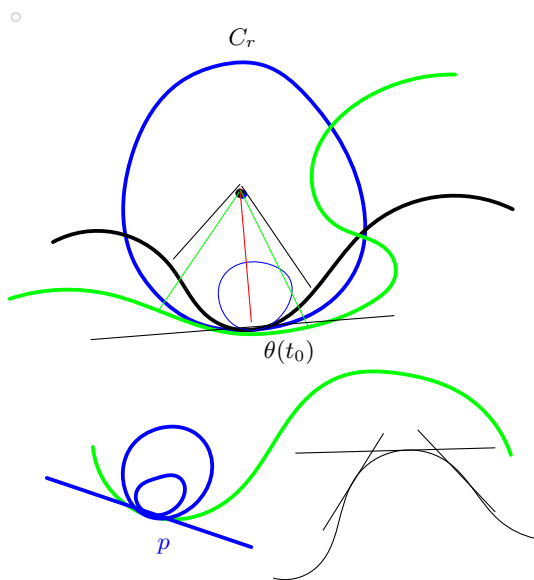
The curvature will be $1/r$ for that r

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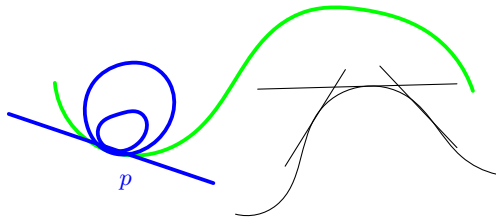
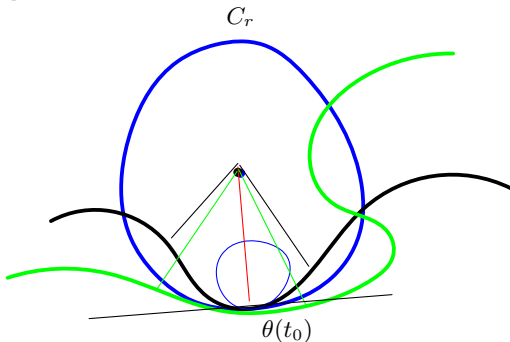
Center of the circle: $\theta(t_0) + r\mathbf{N}(t)$

$d(t) = \|\theta(t) - \theta(t_0) - r\mathbf{N}(t)\|$

$f(t) = (\theta(t) - \theta(t_0) - r\mathbf{N}(t)) \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t))$

$f'(t) =$

○



Another view on curvature

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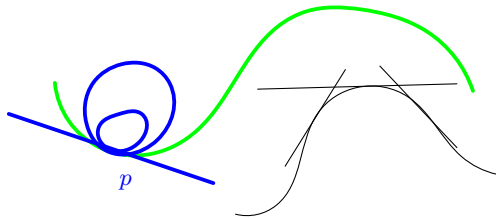
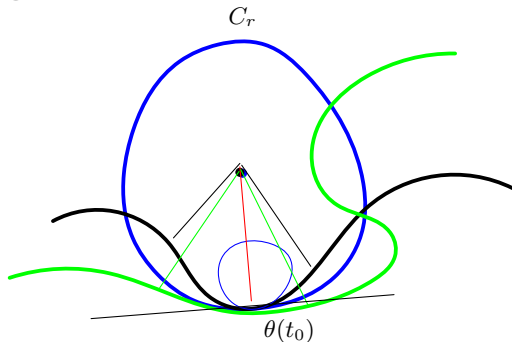
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○



Another view on curvature

Given $\theta : (a, b) \rightarrow \mathbb{R}^2$ is a unit speed parametrization

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Another view on curvature

$$\theta'(t) = \mathbf{T}(t)$$

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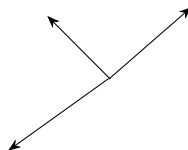
$$f(t) = (\theta(t) - \theta(t_0) - r\mathbf{N}(t_0)) \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t_0))$$

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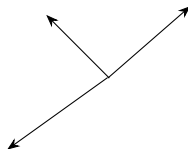
$$f'(t) = 2(\theta'(t) - r\mathbf{N}'(t)) \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t))$$

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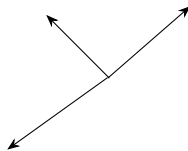
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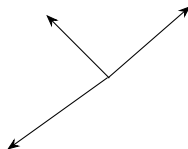
$$\theta'(t) = \mathbf{T}(t)$$

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What is $\mathbf{N}'(t)$?



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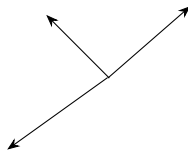
$$\mathbf{N}(t) \cdot \mathbf{N}(t) = 1$$

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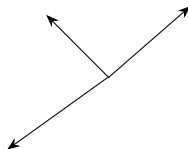
$$2\mathbf{N}(t) \cdot \mathbf{N}'(t) = 0$$

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What is $\mathbf{N}'(t)$?

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$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$



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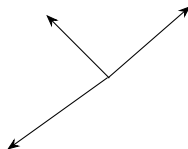
$$\mathbf{T}'(t) = \kappa(t)\mathbf{N}(t)$$

What is $\mathbf{N}'(t)$?

$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

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$$\theta'(t) = \mathbf{T}(t)$$

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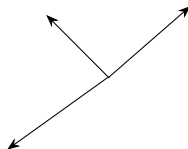
$$\mathbf{T}'(t) = \kappa(t)\mathbf{N}(t)$$

What is $\mathbf{N}'(t)$?

$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

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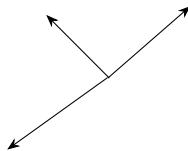
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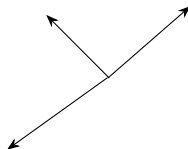
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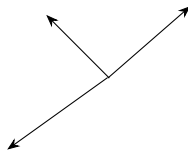
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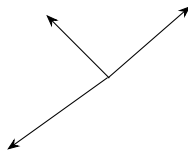
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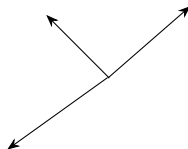
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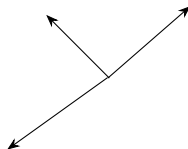
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○

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Conclusion: The best fitting circle to a curve at a point, has a radius which is the reciprocal of the curvature of the curve at that point

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○

$$f'(t) = 2(1 + r\kappa(t))\mathbf{T}(t) \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t))$$

$$f''(t) = (2(1 + r\kappa(t))\mathbf{T}(t))' \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t)) +$$

$$(2(1 + r\kappa(t))\mathbf{T}(t)) \cdot (\theta'(t) - r\mathbf{N}'(t))$$

$$f''(t) = (2(r\kappa'(t)\mathbf{T}(t) + (1 + r\kappa(t))\mathbf{T}'(t))) \cdot (\theta(t) - \theta(t_0) - r\mathbf{N}(t)) +$$

$$(2(1 + r\kappa(t))\mathbf{T}(t)) \cdot (\theta'(t) - r\mathbf{N}'(t))$$

$$f''(t_0) =$$

$$(2(r\kappa'(t_0)\mathbf{T}(t) + (1 + r\kappa(t_0))\kappa(t_0)\mathbf{N}(t_0))) \cdot (-r\mathbf{N}(t_0)) +$$

$$(2(1 + r\kappa(t_0))\mathbf{T}(t_0)) \cdot (\mathbf{T}(t_0) + r\kappa(t_0)\mathbf{T}(t_0))$$

$$f''(t_0) = -2r(1 + r\kappa(t_0))\kappa(t_0) + 2(1 + r\kappa(t_0))^2$$

$$f''(t_0) = 2(1 + r\kappa(t_0))(-r\kappa(t_0) + 1 + r\kappa(t_0))$$

$$= 2(1 + r\kappa(t_0)) = 0$$

$$\kappa(t_0) = -\frac{1}{r}$$

Conclusion: The best fitting circle to a curve at a point, has a radius which is the reciprocal of the curvature of the curve at that point

If we defined curvature as the reciprocal of the radius of the best fitting circle, then it matches with our original definition of curvature.

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